# Sewon Kim

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## Research Interests

Multimodal AI for Autonomous Systems · Large Language Models · Edge Intelligence · Explainable AI

## Education

**Jeonbuk National University** *Jeonju, South Korea*
*Bachelor of Engineering in Computer Science and Engineering; GPA: 4.09/4.5* *Mar. 2022 – Feb. 2026 (Expected)*

- Relevant Coursework: ML/DL, Data Mining, Information Retrieval, Natural Language Processing, Algorithms, Database Systems
- Pursuing micro-degree in Smart Mobility Software (JST Shared University consortium), completion expected Feb. 2026

**Universiti Malaya** *Kuala Lumpur, Malaysia*
*Exchange Student* *Feb. 2023 – Aug. 2023*

## Research Experience

**Undergraduate Researcher** *Feb. 2025 – Present*
*Natural Language Learning Lab (Advisor: Prof. Hyun-je Song)* *Jeonbuk National University*

- Developing knowledge graph-enhanced retrieval-augmented generation (RAG) architectures with PyTorch and LangChain to improve multi-hop reasoning in large language models
- Evaluating trade-offs between retrieval accuracy, latency, and context coherence on multi-hop question answering benchmarks using graph-structured memory representations

**International Research Scholar (IITP-Funded)** *Jun. 2025 – Aug. 2025*
*Korean Software Square (KSW) 2025 Summer Program, Purdue University* *Purdue University*

- Designed and field-deployed LoRa-MQTT IoT sensor network for agricultural monitoring, achieving 110m transmission range with sub-second latency across distributed 8-node architecture (per-node cost: $10–15)
- **Co-authored preprint:** Kim, S., et al., *"WaggleNet: A LoRa and MQTT-Based Monitoring System for Internal and External Beehive Conditions"*, arXiv:2512.07408, 2025. Preparing for conference submission

## Research Projects

**AI-Driven Cultural Heritage Damage Detection and Assessment System** *Jun. 2025 – Nov. 2025*

- Built a cross-platform heritage field-survey app (Flutter + FastAPI + Firebase) that uses a DETR-based model to detect structural damage and output on-site severity grades (A–D) in real-time
- Secured software copyright *"Integrated Survey Registration and Damage Diagnosis Management App"* (Korea Copyright Commission, Reg. No. C-2025-044615); filed patent on the integrated survey and damage-diagnosis workflow (patent pending)

**Hybrid Physical-AI Model for Industrial Emission Impact Prediction** *Feb. 2025 – Jun. 2025*

- Developed hybrid model integrating AERMOD physical diffusion with Transformer-based deep learning for urban air quality prediction using 7 years of TMS (Tele-Monitoring System) time-series data (2018–2024)
- Built automated multi-source data pipeline with LSTM-based imputation; deployed real-time prediction dashboard with 3D spatial visualization using Streamlit and Folium

**YOLOv8-Based Crop Disease Detection for Smart Agriculture** *Sep. 2024 – Dec. 2024*

- Developed YOLOv8 detection system achieving mAP@0.5 of 0.81 for crop disease identification; built large-scale labeled dataset by combining open-source and field-collected images using RoboFlow
- Improved model performance by transitioning from unsupervised Grounding DINO + SAM to supervised YOLOv8 framework; implemented five-level crop health classification based on disease ratio analysis

## Teaching Experience

### Teaching Assistant – Algorithms & Linux Programming
*Sep. 2025 – Present*
*Department of Computer Science and Engineering* — *Jeonbuk National University*

- Mentored 80+ students in algorithm design, 50+ students in Linux programming; developed grading scripts and rubrics to standardize homework and exam evaluation
- Held weekly office hours to support students with data structures, algorithmic problem solving, and Unix/Linux environments (compilation, debugging, shell tools)

## Honors & Awards

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| President's Award for Excellence in Core Competencies (Highest Honor) | *2025* |
| President's Award, Keun-Saram Black Belt Scholarship (Competitive Merit Scholarship, $3,400) | *2025* |
| Excellence Award, Capstone Design Competition (Two-Time Recipient) | *2024 – 2025* |
| Commendation, Big Data Supporters Program, Ministry of Science and ICT, Korea | *2024* |
| President's Award for Outstanding Student, Honors Residential College | *2024* |

## Professional Development

### International Participant, Data Science & AI Summer School
*Jul. 2024*
*Bucharest University of Economic Studies (42-hour intensive program)* — *Predeal, Romania*

- Selected for competitive international program combining lecture-style instruction and hands-on Python/Jupyter implementation covering LLM reasoning and calibration, knowledge graphs (Neo4j), agentic AI frameworks, time-series forecasting, and ML pipeline orchestration

### Data Architecture Semi-Professional (DAsP) Certification
*Oct. 2025*
*Korea Data Agency*

## Leadership & Service

### Student Representative, Honors Residential College
*2022 – Present*
*Jeonbuk National University*

- Organized academic workshops and peer mentoring programs for university residential college students

### Lead Student Representative, Big Data Supporters
*Mar. 2024 – Mar. 2025*
*Big Data Innovation Convergence College, Jeonbuk National University*

- Led student-driven data literacy initiatives and outreach activities promoting data science education

## Technical Skills

**Programming**: Python, C/C++, Java, SQL, Dart
**ML Frameworks**: PyTorch, TensorFlow, LangChain, Hugging Face, OpenCV, scikit-learn
**Development & Tools**: FastAPI, Flask, Flutter, Git, Docker, AWS (EC2, S3), Linux, MQTT
**Research Areas**: Deep Learning, NLP, Computer Vision, Knowledge Graphs, Time-Series Forecasting, IoT Systems